AI IN SPACE AND ORBITAL OPERATIONS - VOLUME 2

The AI Mission Design and Autonomy Playbook

How AI searches the design space, trades the mission, and earns an autonomy envelope that flight can honestly carry.



Gates MD0-MD5 | Eight templates | 20 golden rules

A generated trajectory is not a designed mission. Design AI is licensed per mission, per spacecraft, per orbit regime and per failure mode - never in general, and never on a pretty render.

CHAPTER 1 - THE DESIGN DECISION BRIEF

Design is where a mission is won, lost, or quietly made unflyable

Volume 1 set the licensing rule. This volume takes it upstream, into the rooms where trajectories are chosen, margins are spent and autonomy is promised - years before the spacecraft exists. Design decisions are the most leveraged in any mission and the least reversible: a bad orbit, an undersized battery or an autonomy promise the flight software cannot keep will be paid for on every pass, by operators who were not in the room.

AI changes the economics of that room. It can explore ten thousand design points overnight, rank trades no human team could enumerate, and draft autonomy architectures in hours. It cannot own the consequence. This volume separates what design AI may explore, what it may recommend, and what a named mission designer must still sign.

The brief in one page

- **Exploration is cheap; commitment is not.** Let AI widen the search, then narrow by human judgement on the trades that bind.
- **Every AI-found optimum is a hypothesis** until it survives independent physics, margins and a Monte Carlo it did not tune.
- **Design the autonomy envelope with the mission,** not after it: authority, aborts and safe states are design outputs, not operations patches.
- **Margins are the mission's savings account.** A design that spends them to win a trade study has borrowed from flight.

The design rule: **AI proposes the space of missions; humans choose the one mission, and sign for its worst day, not its best.**

EXECUTIVE LENS

Fund exploration widely and commitment narrowly.

MISSION LENS

Ask what the optimum costs on the worst day, not the nominal day.

OPERATIONS LENS

An unsigned design is a suggestion, however many runs produced it.

Five rungs separate a generated design from a mission baseline

Rung	Name	What it proves	What it does not
1	Generated	The system produces a candidate design, trajectory or architecture	Feasibility against real constraints
2	Explored	The candidate survives the programme's own models and constraint checks	That those models match physics
3	Traded	It has been compared honestly against alternatives, with margins and costs visible	Robustness to dispersion and failure
4	Verified	Independent analysis, high-fidelity simulation and Monte Carlo confirm it inside its envelope	Transfer to another mission, bus or regime
5	Transferable	Re-verified for the new mission class with its own evidence	Nothing transfers by family resemblance

Most AI design claims live at rung 1 or 2 and arrive in the review in rung 4 language: words like optimal, robust and proven attached to outputs nobody independent has checked. The ladder is the correction. Every design artefact in the baseline carries its rung, its evidence and the name of whoever verified it.

Designs also fall down the ladder. A supplier change, a new payload mode, a revised debris environment or a launch-vehicle switch can each drop a verified design back to explored. The design register in Chapter 19 tracks the rung across the programme's life, not just at the preliminary design review.

EXECUTIVE LENS

Ask for the rung in every design review, by artefact.

MISSION LENS

Optimal is a rung 4 word; ban it below that.

OPERATIONS LENS

A change anywhere is a re-verification question everywhere it touches.

Grade every design stage before AI touches any of it



Figure 1 - One spacecraft across its design time planes: concept wireframe, engineering model and flight article, each at its own rung.

Stage	AI today	Grade	First licence
Objectives and requirements drafting	Decomposition, traceability checks	Strong assist	Draft, human owns
Orbit and trajectory search	Wide design-space exploration	Strong, bounded	Propose candidates
Trade studies	Multi-objective ranking, sensitivity	Strong	Rank, human chooses
Subsystem sizing and co-design	Coupled sizing, margin tracking	Assist	Propose, engineer verifies
Autonomy architecture	Function allocation, envelope drafting	Assist	Draft, board signs
Fault management design	Failure-tree drafting, coverage analysis	Assist only	Draft, safety signs
Verification planning	Case generation, Monte Carlo design	Strong	Generate, independent check
Operations concept drafting	ConOps, procedure skeletons	Assist	Draft, operators own
Cost and schedule estimation	Analogy and parametric models	Assist, optimistic bias	Advise, with uplift
Design documentation	Drafting, consistency checking	Strong	Draft, author signs

The pattern matches Volume 1: AI is strongest where evaluation is cheap and reversible - search, ranking, drafting, checking - and weakest where the artefact commits mass, propellant, authority or safety. Licence stage by stage, and never let a strong grade in search launder a weak one in fault management.

EXECUTIVE LENS

The map, not the demo, sets the programme's AI scope.

MISSION LENS

Search grades never transfer to safety grades.

OPERATIONS LENS

Let the cheap stages fund discipline in the expensive ones.

AI can draft the mission; it cannot decide what the mission is for

Every design failure in this volume traces back, eventually, to an objective that was vague, a requirement that was untestable, or a success criterion quietly rewritten mid-programme. AI drafting tools make all three easier to produce at scale - fluent, plausible, untraceable requirements in unlimited quantity.

The requirements discipline

- **Objectives are human-signed before any search begins.** The optimiser optimises what it is given; a wrong objective, efficiently pursued, is the most expensive output in design.
- **Every requirement carries its verification method** - test, analysis, inspection or demonstration - at the moment it is written. A requirement no one can verify is a hope with a number.
- **AI checks traceability and consistency;** humans own intent. Contradiction-finding across a thousand requirements is exactly where machines earn their place.
- **Derived requirements from AI tools are flagged as derived,** with their parent and their author, so a model's inference never silently becomes programme law.

The test of this chapter is simple: at the critical design review, can every requirement on the wall name its parent objective, its verifier and its rung? If not, the design AI has been writing fiction in the programme's own format.

EXECUTIVE LENS

Sign the objective before you fund the search.

MISSION LENS

Untestable requirements are debts issued against flight.

OPERATIONS LENS

Machines check consistency; only people can mean something.

The widest search still answers to orbital mechanics

Trajectory design is AI's showcase stage: the physics is well understood, the objective functions are explicit, and the search space is far larger than any human team can walk. Optimisers find transfer windows, phasing plans and station-keeping strategies that manual design misses, and they find them in hours.

The risks are quieter. An optimiser exploits every looseness in its model: a simplified gravity field, an optimistic thruster, drag that behaves better in the model than in the thermosphere. The result is a trajectory that is optimal for a planet that does not quite exist.

Licence conditions for trajectory AI

- **Two-model rule:** any AI-proposed trajectory is re-propagated in an independent, higher-fidelity model before it enters a trade.
- **Dispersion before admiration:** the candidate is shown with its dispersions - injection error, missed-thrust, drag uncertainty - not as a single elegant line.
- **Propellant honesty:** delta-v budgets carry the programme's margin policy unmodified; an optimiser may not spend margin to improve its own score.
- **Constraint provenance:** keep-out zones, conjunction policy and disposal requirements are inputs signed by their owners, never defaults the tool chose.

EXECUTIVE LENS

Celebrate the search; verify on a second model, always.

MISSION LENS

A trajectory without dispersions is marketing.

OPERATIONS LENS

Margin policy outranks the optimiser's objective function.

A ranked list is not a decision, and a Pareto front is not a verdict

Multi-objective search gives design teams something genuinely new: the shape of the possible. A Pareto front showing mass against coverage against cost against risk lets a board see what it is actually choosing between, instead of choosing between two teams' favourite point designs.

It also creates a new failure: the front is only as honest as its objectives and constraints. Leave resilience out of the objective set and the front will efficiently eliminate it. Weight cost on development only and the front will select designs that are ruinous to operate for fifteen years.

Running an honest trade

- **Objectives and weights are signed before the run,** and any change re-runs the trade in the open, with the change logged as a decision.
- **Show the near-misses.** The designs just off the front - slightly heavier, far more operable - are often the right mission; AI must surface them, not bury them.
- **Sensitivity is part of the answer.** A choice that flips when an assumption moves 5 percent is not a choice yet; it is a research task.
- **The decision record names what was traded away,** in plain language, so the future operator reading it in year nine knows what the board knowingly gave up.

EXECUTIVE LENS

Sign the weights before the run, or the run will sign them for you.

MISSION LENS

The near-miss designs are where judgement lives.

OPERATIONS LENS

Record what was traded away; flight will ask.

Coupled optimisation spends margins jointly - and can hide the spending

Spacecraft subsystems are coupled: power feeds thermal, thermal bounds payload duty, structure carries everything. AI co-design tools optimise across the couplings and find integrated designs lighter than any sequential, subsystem-by-subsystem pass. That is real value, and it carries the volume's subtlest risk.

Sequential design wastes mass visibly. Coupled design can consume margin invisibly, shaving every subsystem's reserve simultaneously because the joint model says the combination works - on a nominal day, with nominal dispersions, in the model's weather. The design is optimal and brittle in the same spreadsheet.

Margin rules under co-design

- **Margins are held centrally,** owned by the chief engineer, and no optimiser may allocate, borrow or net them across subsystems.
- **Every co-designed point reports margin by subsystem** alongside the headline mass, so erosion is visible in the same table that celebrates the saving.
- **Worst-case coupling cases are mandatory runs:** hot case with degraded arrays, cold case with a failed heater, end-of-life everything, together.
- **Growth allowance follows heritage,** not optimism: novel subsystems carry novel-subsystem allowances whatever the model believes.

EXECUTIVE LENS

Joint optimisation, central margins - never the reverse.

MISSION LENS

If the saving table hides the margin table, reject both.

OPERATIONS LENS

Brittle and optimal are the same design on different days.

Allocate every function to a level of authority before anyone builds it

Autonomy architecture is the design discipline this volume exists to teach: deciding, function by function, what the spacecraft may sense, decide and do alone; what it may propose; what it must ask; and what it must never touch. Volume 1's authority levels - A1 advise, A2 propose, A3 act in envelope, A4 act across a fleet - become, here, design artefacts with the same status as the structure.

Design output	Contents	Signed by
Function allocation matrix	Every function, its authority level, its rationale	Mission director
Envelope specification	Bounds, aborts, expiries per A3/A4 function (Chapter 9)	Flight director
Mode and state design	Modes, transitions, and the safe state each mode can reach alone	Chief engineer
Ground override design	How the ground halts, demotes or reconfigures each function	Operations lead
Degradation plan	What the mission becomes when autonomy is switched off entirely	Mission director

The degradation plan is the test of the whole architecture. A mission that cannot be flown, at reduced capability, with its autonomy off is not an autonomous mission; it is a hostage arrangement with a model.

EXECUTIVE LENS

Architecture first: authority is a design output, not a feature.

MISSION LENS

If it cannot fly with autonomy off, it does not fly.

OPERATIONS LENS

Every A3 function names its halt before it names its benefit.

The envelope is the licence, drawn in advance and in numbers



Figure 2 - The signed autonomy envelope: the spacecraft acts alone inside the amber boundary, and the red path outside it belongs to humans.

An envelope is the set of conditions inside which an autonomous function may act without asking: state bounds, resource floors, timing windows, environmental limits and abort triggers, each with a value, an owner and an expiry. Envelope design is where autonomy stops being a capability claim and becomes a contract.

- **Bounds from physics and policy, not from performance.** The envelope edge sits where safety and mission rules say, with margin inside it - never at the edge of what the model once achieved in simulation.
- **Aborts are designed, rehearsed and counted.** Every envelope names the abort it triggers on exit, the safe state that abort reaches, and the telemetry that will prove it happened.
- **Envelopes expire.** Ageing, radiation, a changed debris field and software updates all shrink what yesterday's evidence supports; expiry dates force the renewal conversation in design, not during an anomaly.
- **No self-amendment.** A function may recommend an envelope change; only a signed design change grants one. Self-widening autonomy is grounded on discovery, per Volume 1's standing refusals.

EXECUTIVE LENS

Draw the envelope in numbers, or it will be drawn by an incident.

MISSION LENS

Abort design is half the envelope.

OPERATIONS LENS

Expiry is a feature: evidence ages in orbit.

Put each decision where the light-time, the data and the consequence say it belongs

The oldest allocation question in mission design is newly sharp: with capable onboard models, what still belongs on the ground? The answer is a function of three clocks - the decision deadline, the contact schedule, and the light-time delay - and one asymmetry: the ground has people, compute and time; the spacecraft has presence, immediacy and no help.

Decision class	Belongs	Why
Milliseconds to seconds: pointing, rate control, immediate safing	Onboard, deterministic	No link exists at that clock speed
Minutes: anomaly first response, conjunction inside a contact gap	Onboard, inside envelope	Deadline falls between passes
Hours: re-planning, trade-offs, campaign changes	Ground, AI-assisted	Time exists; use the bigger brain
Days: strategy, envelope changes, software uploads	Ground, human-signed	Consequence is programme-level
Deep-space cruise decisions inside long light-time	Onboard, pre-committed plans	Ground advice arrives after the event

Design error runs both ways: ground-romantic designs that strand a spacecraft helpless in a contact gap, and autonomy-romantic designs that fly decisions the ground could have made better, slower and reversibly. The allocation table, signed at design time, is the defence against both romances.

EXECUTIVE LENS

Allocate by clock and consequence, not by fashion.

MISSION LENS

Presence onboard, judgement on the ground, wherever time allows.

OPERATIONS LENS

Write the no-contact decision into the design, not the anomaly report.

A twin is evidence only to the extent that its differences from flight are known

Every claim in this volume is mediated by a model of a spacecraft that does not yet exist, flying through an environment that can only be approximated. AI design tools raise the stakes: they search so widely that they reliably find the places where the twin and reality disagree - and present those places as optima.

Fidelity discipline

- **Every twin carries a fidelity statement:** what it models, what it simplifies, what it ignores, and the test data each rests on. A twin without one is an illustration.
- **Anchoring is scheduled, not hoped for:** component tests, engineering-model runs and, later, early flight telemetry are planned events that re-anchor the twin at each phase.
- **Optimisers are fenced from known twin weaknesses.** Where the twin is soft - ageing, slosh, flexible modes, thermosphere weather - design constraints hold the search away from the soft ground.
- **Twin and flight software are separate artefacts,** independently maintained; a design verified only against its own twin has been verified against itself.

After launch the twin earns a second life as the operations mirror of Volume 4, but only if design kept its fidelity statement honest. Programmes inherit the twin they documented, not the one they remember.

EXECUTIVE LENS

Publish the twin's ignorances with its predictions.

MISSION LENS

An optimiser will find your model's errors; fence them first.

OPERATIONS LENS

Verified against its own twin means unverified.

Robustness is a distribution, and the tails are the mission

Nominal performance is a press release. Missions live in the tails: the cold battery, the late burn, the sensor at end of life in a bad season, together. AI transforms verification economics - Monte Carlo campaigns of tens of thousands of dispersed runs, automated case generation hunting for failure combinations, and coverage analysis that shows which corners of the envelope no run has visited.

- **Dispersions are owned by discipline engineers,** not tuned by whoever wants the run to pass. A Monte Carlo with friendly dispersions is a longer, more expensive fiction.
- **Failure-seeking is a separate campaign** from performance-measuring: adversarial case generation is rewarded for breaking the design, and its finds are design inputs, not embarrassments.
- **Coverage is reported by envelope region,** so the board sees which corners remain unvisited before it signs, and accepts them knowingly or not at all.
- **Independence holds:** the verifying models, data and team are not the designing ones. AI may generate the cases; it may not grade its own design's homework.

Sign-off standard: **the design holds across the dispersed distribution, inside the envelope, with margins intact - or it is not yet a design.**

EXECUTIVE LENS

Judge the tails; the nominal case will look after itself.

MISSION LENS

Pay one campaign to break what the other campaign built.

OPERATIONS LENS

Unvisited corners are signed risks or open work - never silence.

Design the worst day first: detection, isolation, recovery and the road back

Fault detection, isolation and recovery - FDIR - is where autonomy stops being an efficiency and becomes the mission's survival system. AI contributes genuinely: failure-tree drafting from the design model, coverage analysis that finds the fault no sensor can see, and response plans checked against every mode and state.

It also tempts a category error: learned anomaly detection as the primary safety net. A learned detector is a statistical instrument with unknown coverage on faults it has never seen - acceptable for triage and early warning, never as the sole protection against a catastrophic fault. Hard protections stay deterministic, simple and analysable; learned systems advise around them.

FDIR design rules

- **Every catastrophic fault has a deterministic detection and a safing path** that no model can inhibit, delay or reinterpret.
- **Recovery is designed per mode:** what recovers automatically, what waits for the ground, and what is never attempted autonomously are three separate, signed lists.
- **False-safing is costed.** A system that safes weekly will be inhibited by its operators within a year; availability is a safety property, because defeated protections protect nothing.
- **The FDIR design is tested by injection,** in simulation and on the engineering model, with the operations team - not the design team - holding the stopwatch.

EXECUTIVE LENS

Deterministic for survival, learned for warning - never merged.

MISSION LENS

A protection operators defeat is a design failure.

OPERATIONS LENS

Worst day first is a scheduling rule, not a slogan.

Reviews, overrides and the designer who must be able to say no

Design AI fails socially before it fails technically. A trade tool whose outputs arrive as finished answers trains its engineers out of the judgement the programme will need at the critical design review and, later, at 3 a.m. over the actual spacecraft. Teaming is therefore designed, like everything else in this volume.

- **AI shows its working as a default:** assumptions, sensitivities and rejected alternatives are part of the output object, not a favour the tool does when asked.
- **Reviews sample the reasoning, not the summary.** Each design gate independently re-derives a sample of AI-produced results by classical methods, and the sample rate is published.
- **Override is a first-class outcome.** A designer rejecting an AI recommendation records why; override patterns are programme data that expose both weak models and weak reviewers.
- **Skill is maintained deliberately:** rotation through manual design work, simulator exercises without assistance, and early-career engineers doing the fundamentals before the tools. The programme's licence to use design AI rests on retaining people who could do without it.

EXECUTIVE LENS

If reviewers cannot re-derive it, they cannot sign it.

MISSION LENS

Record overrides; they are the audit trail of judgement.

OPERATIONS LENS

Keep the skill the tool replaced, or lose the right to use the tool.

Operability is designed in or paid for on every pass for fifteen years

The cheapest operations decision is the one taken in design: telemetry that answers the questions flight will actually ask, command granularity that matches how anomalies actually unfold, and autonomy whose behaviour the night shift can predict, explain and bound.

- **Operators sit in design reviews from phase A,** with authority to raise operability requirements to the same level as performance ones - not as reviewers of a finished design, but as co-authors of it.
- **Every autonomous function ships with its explanation surface:** what it saw, what it decided, what it nearly did, in telemetry the ground can actually downlink and read.
- **Design for the anomaly, staff for the nominal:** contact plans, safing behaviour and ground tooling are sized around the bad week, and the staffing model survives the chief operator's holiday.
- **End of life is a design case:** disposal, passivation and the final commands are designed, verified and rehearsed like launch - because a derelict is a fifteen-year operations outcome chosen in a design meeting.

Volume 3 takes up operations in full. This chapter's claim is narrower: by the time operations inherits the spacecraft, the expensive decisions have already been made, and the kindest thing design can hand over is a spacecraft whose autonomy is legible.

EXECUTIVE LENS

Operators as co-authors, not reviewers of last resort.

MISSION LENS

Legible autonomy is an operability requirement.

OPERATIONS LENS

Disposal is a design case with a launch's discipline.

The design is an attack surface: models, data, tools and parts

A mission designed with AI inherits the supply chain of its tools: training and heritage data of unknown provenance, design models that encode the spacecraft's deepest details, third-party components and libraries, and the ground toolchain that will one day command the fleet. Adversaries read design-phase artefacts with more patience than flight telemetry.

- **Design models are crown jewels:** the full digital twin, its fault trees and its envelope specifications are classified, access-logged and segmented like flight keys.
- **Tool provenance is checked like part provenance:** where the model was trained, on whose data, with what right, and whether programme data fed it - with contractual answers, not assurances.
- **Programme data does not train vendor models by default;** any such use is a separate, priced, revocable agreement, as Volume 1's worked example enforced.
- **Command paths are designed adversarially from day one:** authentication, independent command checking and upload authorisation are architecture, not a security review bolted on before launch.

Volume 9 covers space security and assurance in full. The design-phase rule stands alone: the cheapest time to keep an adversary out of a mission is before the mission exists.

EXECUTIVE LENS

Guard the twin like the keys - it is worth more.

MISSION LENS

Contractual provenance, or no tool on programme data.

OPERATIONS LENS

Security in the architecture, not in a pre-launch review.

Who owns the mission you designed with someone else's model

Design AI concentrates intellectual property questions that procurement usually meets too late. The design was produced with a vendor's model, trained partly on heritage data, tuned on programme problems. Who may reuse it, who may walk away with it, and what happens to the programme's ability to verify its own design when the licence lapses?

Question	Standing answer in this series
May the vendor train on programme design data?	Only under a separate, priced, revocable agreement
Who owns AI-generated design artefacts?	The programme, with full working: assumptions, cases, rerun rights
Can the programme re-verify without the vendor?	Must be demonstrable before baseline, not promised for later
What if the model version changes mid-programme?	Treated as a design change: affected artefacts return a rung for re-verification
Export and jurisdiction limits on tools and data?	Mapped per programme at contract, reviewed at every gate

The re-verification test is the sharpest: if the vendor vanished the day after the critical design review, could the programme still prove its design? If the honest answer is no, the programme does not have a design; it has a subscription with a launch date.

EXECUTIVE LENS

Own the working, not just the drawings.

MISSION LENS

Re-verification without the vendor is demonstrated, not promised.

OPERATIONS LENS

A model update mid-programme is a design change, full stop.

Cost per verified design outcome, in claimed, released and banked hours

Design AI is sold in hours saved across engineering teams. Programmes in this series count differently, in the three-figure discipline of Volume 1: hours claimed by the vendor, hours released from actual tasks, and hours banked - reassigned or removed with the work genuinely gone. Design adds a fourth figure the others do not have: verification cost, because every AI-saved design hour purchases checking hours, and honest accounting nets them.

Measure	Definition	Design trap
Cost per verified design outcome	Fully loaded design plus verification cost, per artefact reaching rung 4	Counting generated artefacts makes fiction look cheap
Rework rate	Share of AI-produced artefacts returned a rung at review or later	Deferred rework lands in integration, at ten times the price
Operations cost consequence	Modelled lifetime ops cost of the chosen design vs its near-misses	Development savings that buy fifteen expensive years
Margin-adjusted value	Saving stated after restoring any margin the design spent to achieve it	Mass saved by spending reserve is borrowing, not saving

The headline rule: a design tool that halves drafting time and doubles review findings has not saved anything yet. Judge it at the gate, on verified outcomes, with the operations lifetime in the same ledger.

EXECUTIVE LENS

Net the verification cost, or the saving is staged.

MISSION LENS

Rework found in integration costs ten times rework found in design.

OPERATIONS LENS

Count the fifteen operating years in the design ledger.

One register, one chief engineer, and gates that can stop the programme

Design risk in an AI programme is a portfolio: model risk in the tools, fidelity risk in the twins, authority risk in the autonomy architecture, and the oldest risk of all - a schedule that eats verification first because verification complains last.

- **The design register** holds every design function, artefact class and autonomy allocation with its ladder rung, evidence, verifier and expiry, reviewed at every gate and after every major change.
- **One accountable chief engineer** owns margin policy, envelope policy and the register; committees advise, a named human signs.
- **Gates can stop the programme.** A gate that has never stopped anything is a ceremony. MD gates in this volume carry explicit stop authority, exercised in the worked example that follows.
- **Change is the enemy state.** Late changes are re-priced in rung terms - what falls, what must be re-verified, what it costs - before they are approved, so the board sees the true invoice.
- **Anyone can halt a design release,** as anyone can halt an autonomous function in flight. Halts are thanked, analysed and recorded; a programme that punishes halts will receive silence instead, at flight prices.

EXECUTIVE LENS

One register, one signer, gates with teeth.

MISSION LENS

Price a change in rungs before approving it in money.

OPERATIONS LENS

Protect the halt in design as fiercely as in flight.

WORKED EXAMPLE - AURORA ORBITAL (ILLUSTRATIVE)

A vendor design claim for Aurora's next generation, audited down to verified truth

Aurora Orbital is fictional. All figures are illustrative and labelled as such wherever they are used.

Aurora, the 24-satellite Earth-observation operator of Volume 1, commissions its second-generation spacecraft. A design-AI vendor claims a 46 percent design-cost saving and 38,000 engineering hours across requirements drafting, trajectory and trade search, co-design and verification planning, with the resulting design described as verified and its autonomy architecture ready to baseline.

Measure	Vendor claim	Audited (illustrative)
Design saving	46%	18% verified
Engineering hours	38,000 claimed	19,400 released; 12,100 banked
Trajectory design	Verified	Rung 4 confirmed after independent re-propagation
Trade study	Optimal design selected	Rung 3: resilience absent from the objective set, re-run ordered
Co-design	11% mass saving	Rung 2: saving achieved by netting margins across subsystems
Autonomy architecture	Ready to baseline	Rung 2: degradation plan missing; mission unflyable with autonomy off

The audit found Volume 1's pattern one phase earlier: the search work was genuinely strong, the commitment language ran two rungs ahead of the evidence, and the mass saving was margin wearing a costume.

EXECUTIVE LENS

Audit per artefact class; programme-wide language hides weak cells.

MISSION LENS

Released and banked are the only hours that fund anything.

OPERATIONS LENS

The prettiest number in the pack was borrowed margin.

WORKED EXAMPLE - DECISIONS (ILLUSTRATIVE)

Licensed in part, refused in part, and the refusals are the system working

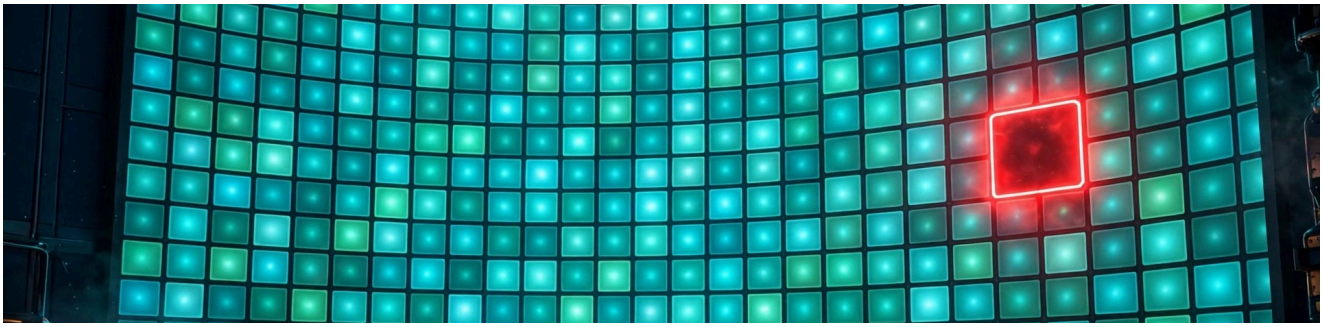


Figure 3 - The Aurora design operations wall (illustrative): design and autonomy cells licensed one by one, with the co-design cell held in red.

Licensed at MD4

- Trajectory and orbit search as a proposing tool, under the two-model rule, with dispersions mandatory in every candidate pack.
- Requirements drafting and traceability checking, with derived requirements flagged and human-owned.
- Verification case generation, with dispersions owned by discipline engineers and an independent failure-seeking campaign.

Refused or deferred

- Co-design mass saving as presented: refused; the design returns with margins held centrally, accepting 4 percent verified mass saving instead of 11 percent borrowed.
- Autonomy architecture baselining: deferred until the degradation plan demonstrates the mission flown, at reduced capability, with autonomy off.
- Trade conclusion: suspended; the trade re-runs with resilience and lifetime operations cost in the signed objective set.

Cost per verified design outcome in the licensed cells: 0.82 against a 0.98 pre-AI baseline (illustrative). MD5 scale was granted in part: search, drafting and verification support scale to the follow-on mission study; co-design and autonomy baselining scale only when their re-submitted evidence reaches rung 4.

EXECUTIVE LENS

Grant in part is a success state, not a compromise.

MISSION LENS

A refused bundle can return as an evidenced application.

OPERATIONS LENS

Scale only what reached rung 4; the rest is a new claim.

THE MISSION DESIGN STANDARD

The one-page standard every design programme in this series signs

- Design AI is licensed per mission, per spacecraft, per orbit regime and per failure mode.
- Every design artefact carries its ladder rung, its evidence and its verifier.
- Objectives and trade weights are human-signed before any search runs.
- AI-proposed trajectories are re-propagated on an independent model before entering a trade.
- Margins are held centrally; no optimiser nets or borrows them.
- Authority levels A1 to A4 are allocated function by function, in a signed matrix.
- Every A3 or A4 function has a designed envelope, an abort and a safe state.
- Every mission has a degradation plan: flyable, at reduced capability, with autonomy off.
- Catastrophic faults are protected deterministically; learned detection advises, never replaces.
- Verification is independent, dispersed and failure-seeking; AI never grades its own design.
- Programme design data trains no vendor model without a separate, priced, revocable agreement.
- Anyone can halt a design release; halts are thanked, analysed and recorded.

EXECUTIVE LENS

The standard fits one page so it can survive one hard review.

MISSION LENS

If a practice cannot be stated simply, it will not survive a schedule slip.

OPERATIONS LENS

Sign it at programme level; enforce it at artefact level.

FAILURE PATTERNS

Six ways design AI loses missions before they launch

#	Pattern	Signature	Correction
1	The planet that does not exist	Optimum sits exactly on a known twin weakness	Fence the search; two-model rule
2	Margin in a costume	Mass or power saving with margins quietly netted	Central margins; savings stated margin-adjusted
3	The missing objective	Resilience or operability absent from the trade weights	Signed objective set, re-run in the open
4	Nominal-day design	Verification on undispersed, friendly cases	Tail-first Monte Carlo, independent dispersions
5	Hostage autonomy	Mission cannot be flown with autonomy switched off	Degradation plan as a gate artefact
6	The unverifiable baseline	Only the vendor's tool can reproduce the design evidence	Re-verification test before baseline

Each pattern was survivable at the review where it was cheapest, and each was usually visible in the documents of that review. The patterns persist not because teams cannot see them, but because the schedule rewards not seeing them. Gate discipline is the counterweight: a gate that can stop the programme makes seeing them affordable.

EXECUTIVE LENS

Patterns are schedule products; fix incentives, not just checklists.

MISSION LENS

Every pattern was cheapest at one specific review. Name it.

OPERATIONS LENS

A gate that never stops anything teaches teams to hide things.

ASSURANCE QUESTIONS

30 questions every mission design board answers in writing

- 1. What rung is each baselined artefact on, and who verified it?
- 2. Which objectives and weights were signed before the search?
- 3. What was traded away, in plain language?
- 4. Which near-miss designs were surfaced, and why were they passed over?
- 5. Where does the chosen design sit in its dispersed distribution?
- 6. What are the known weaknesses of the twin, and how is the search fenced from them?
- 7. Which trajectories were re-propagated independently?
- 8. What margins remain, by subsystem, after all savings?
- 9. Which margins did any tool propose spending, and who refused?
- 10. What is the function allocation matrix, and who signed it?
- 11. For each A3 function: bounds, abort, safe state, expiry?
- 12. Can the mission fly, reduced, with autonomy off? Demonstrated how?
- 13. Which decisions fall in contact gaps, and who owns them by design?
- 14. Which catastrophic faults rely on learned detection alone? (Required answer: none.)
- 15. What is the false-safing rate, and who costed it?
- 16. What share of the envelope has Monte Carlo actually visited?
- 17. Who owned the dispersions, and who tried to soften them?
- 18. What did the failure-seeking campaign break?
- 19. Can operators explain what each autonomous function will do next?
- 20. Is disposal designed and rehearsed to launch standard?
- 21. Which programme data has any vendor trained on, under what agreement?
- 22. Could the programme re-verify its design if the vendor vanished tomorrow?
- 23. What happens to every rung when the next major change lands?
- 24. What is the cost per verified design outcome, net of verification?
- 25. What lifetime operations cost did the chosen design accept?
- 26. Which requirements are derived, and who owns their intent?
- 27. What sample of AI work was independently re-derived at this gate?
- 28. Who has halted a design release, and what happened next?
- 29. Which envelopes expire before launch, and what renews them?
- 30. What would make this board refuse its own design?

EXECUTIVE LENS

Written answers, or the questions were decoration.

MISSION LENS

Question 14 has a required answer; keep it that way.

OPERATIONS LENS

Question 30 is the board examining itself. Do not skip it.

THE GATES

MD0-MD5: the six gates from intent to licensed design

Gate	Name	Entry evidence	Exit decision
MD0	Intent	Signed objectives, exclusions, authority policy	Proceed to exploration
MD1	Exploration	Design map graded, tools licensed per stage, fidelity statements	Search may run
MD2	Trades	Signed objective set, Pareto evidence, near-misses surfaced	Point design chosen, trades-away recorded
MD3	Architecture	Allocation matrix, envelopes, FDIR and degradation plans	Autonomy architecture baselined in part or deferred
MD4	Verification	Independent re-propagation, dispersed Monte Carlo, injection tests	Design licensed per artefact class
MD5	Scale and transfer	Rung 4 evidence per class, re-verification without vendor demonstrated	Scale in part, transfer per mission, or refuse

Gates license classes of artefact and classes of authority, never the programme as a whole. A mission can pass MD4 for its trajectory design and be deferred at the same gate for its autonomy architecture, as Aurora was - and that asymmetry is the gate system working, not failing.

EXECUTIVE LENS

Partial passes are the normal, healthy output.

MISSION LENS

Stop authority is written into each gate, in advance.

OPERATIONS LENS

Transfer is a fresh gate, not a family discount.

FIRST 90 DAYS

A design programme resets its AI practice in one quarter

Days	Move	Done when
1-15	Grade the design map; inventory every AI tool in use, registered or not	Map signed; unregistered tools suspended from baseline work
10-30	Stand up the design register with rung, evidence, verifier and expiry per artefact class	Register reviewed by the chief engineer
20-45	Centralise margin ownership; re-state all claimed savings margin-adjusted	Margin table in every design pack
30-60	Draft allocation matrix, envelopes and degradation plan for the lead mission	Reviewed with operations in the room
45-75	Run the two-model rule and one failure-seeking campaign on the current baseline	Finds logged as design inputs
60-90	Execute vendor and data agreements; run the re-verification-without-vendor test	Test result to the board, pass or priced remediation

Ninety days does not finish a design. It establishes the register, the margins and the gates - the machinery that makes every later claim checkable - and it produces the first honest set of numbers the board has seen.

EXECUTIVE LENS

Machinery first: register, margins, gates.

MISSION LENS

The first honest numbers are the quarter's real deliverable.

OPERATIONS LENS

Suspend unregistered tools from baseline work, kindly and immediately.

What a design licence rests on, and how sectors differ

The evidence pack

- Signed objectives, trade weights and the decision record, including what was traded away.
- Design register extract: rung, evidence, verifier and expiry per artefact class.
- Fidelity statements for every twin and model used, with anchoring test results.
- Independent verification reports: re-propagation, dispersed Monte Carlo, injection tests.
- Allocation matrix, envelope specifications, FDIR analysis and degradation plan.
- Vendor, data and IP agreements, with the re-verification test result.

Sector annex

Sector	Design emphasis
Earth observation constellations	Revisit and coverage trades, downlink-first design, fleet allocation
Communications	Capacity and interference trades, spectrum constraints as signed inputs
Navigation and timing	Integrity design first; autonomy never inside the integrity path
Science and deep space	Light-time allocation, pre-committed plans, frozen critical-phase design
Crewed missions	Advisory ceilings on safety-of-life functions, abort design owned by humans
Servicing and assembly	Hold points, proximity envelopes, two-party consent in the architecture

EXECUTIVE LENS

No pack, no licence - the pack is the licence's body.

MISSION LENS

Integrity paths stay deterministic in every sector.

OPERATIONS LENS

The annex adjusts emphasis, never the standard.

TEMPLATES 1-4

Fill-in templates for mission design and autonomy - part one

Template 1 - Design AI decision brief

Mission / decision / artefacts in scope / stages licensed / exclusions signed / signer and date.

Template 2 - Claim and rung register entry

Artefact / claim as stated / rung / evidence / verifier / expiry / change triggers that return it a rung.

Template 3 - Trade study record

Signed objectives and weights / runs performed / Pareto summary / near-misses surfaced / decision / what was traded away, in plain language.

Template 4 - Autonomy envelope specification

Function / authority level / state bounds / resource floors / timing windows / abort trigger and safe state / owner / expiry.

Each template is completed in the programme's own system and filed with the evidence pack. A blank field is a decision not yet taken, and the gate treats it as such.

EXECUTIVE LENS

Templates are short on purpose: the thinking is the work.

MISSION LENS

Blank fields are open decisions, named as such at gates.

OPERATIONS LENS

File with the evidence pack, not in a chat thread.

TEMPLATES 5-8

Fill-in templates for mission design and autonomy - part two

Template 5 - Function allocation matrix entry

Function / onboard or ground / authority level A1-A4 / rationale / halt method / degradation behaviour.

Template 6 - Degradation plan

Mission with autonomy off / retained capability / lost capability accepted / crew and operator load / demonstration evidence.

Template 7 - Verification campaign plan

Cases and dispersions with owners / failure-seeking scope / envelope coverage target / independent checker / pass standard.

Template 8 - Gate sign-off MD0-MD5

Gate / artefact classes licensed / classes deferred or refused with reasons / conditions / re-submission evidence required / signers.

Used together, the eight templates carry a mission from intent to licensed design with no claim travelling without its evidence attached.

EXECUTIVE LENS

Eight templates, one chain from intent to licence.

MISSION LENS

Refusals are recorded on the same form as approvals.

OPERATIONS LENS

Re-submission evidence is named at the gate, not negotiated later.

GOLDEN RULES

20 golden rules of AI mission design and autonomy

- 1. AI proposes the space; humans choose the mission.
- 2. Optimal is a rung 4 word.
- 3. Sign the objective before the search.
- 4. Two models or no trade.
- 5. Margins are held centrally, always.
- 6. A saving that spends margin is borrowing.
- 7. Show dispersions, not lines.
- 8. Authority is allocated function by function.
- 9. Every envelope names its abort.
- 10. No self-amending autonomy.
- 11. Flyable with autonomy off, or not flyable.
- 12. Deterministic for survival, learned for warning.
- 13. Verify independently; never self-grade.
- 14. Design the worst day first.
- 15. Operators co-author, never just review.
- 16. Disposal is a design case.
- 17. Own the working, not just the drawings.
- 18. Re-verify without the vendor before you baseline.
- 19. Record what was traded away.
- 20. Anyone can halt a design release.

EXECUTIVE LENS

Rules 5 and 6 retire more bad designs than any review.

MISSION LENS

Rule 11 is the architecture exam. Set it early.

OPERATIONS LENS

Post the rules in the design room, not the boardroom.

SERIES INDEX

AI in Space and Orbital Operations - the ten volumes

Vol	Title	Gates	Status
1	Space Strategy	SS0-SS5	Built
2	Mission Design and Autonomy	MD0-MD5	Built - this volume
3	Satellite and Constellation Operations	SO0-SO5	Next
4	Ground Systems and Mission Control	GS0-GS5	Planned
5	Earth Observation and Space Data	EO0-EO5	Planned
6	Navigation, Timing and Debris Avoidance	NT0-NT5	Planned
7	Launch, Rendezvous and Servicing	LS0-LS5	Planned
8	Lunar and Deep-Space Operations	LD0-LD5	Planned
9	Space Safety, Security and Assurance	SA0-SA5	Planned
10	Space Capstone and Assurance	CP0-CP5	Planned

Volume 1 set the licensing rule. This volume takes it into the design room: exploration widened by AI, commitment narrowed by signature, and autonomy drawn as an envelope before it is built as a feature. Volume 3, Satellite and Constellation Operations, takes the licensed design into flight.

EXECUTIVE LENS

Two volumes, one register, one licensing rule.

MISSION LENS

Design signs; operations inherits. Keep the chain legible.

OPERATIONS LENS

Volume 3 begins where this one hands over.